

RAJ PATEL

+1(669)302-5489 | raj.patel190702@gmail.com | Boston, MA | [Linkdin](#) | [GitHub](#) | [Leetcode](#)

EDUCATION

University of the Pacific

Aug 2022- Dec 2024

Master of Science in Computer Science

LDRP Institute of Technology and Research

Aug 2017- May 2021

Bachelor of Engineering in Computer Engineering

SKILLS

- **Programming & Frameworks:** Python, C++, Java, C, SQL, JavaScript
- **Python Libraries:** NumPy, Pandas, Matplotlib, PyTorch, TensorFlow, NLTK, XGBoost, Scikit-Learn, Keras
- **ML & DL Architectures:** Linear Regression, Logistic Regression Clustering, SVM, Naïve Bayes, Random Forest, NLP, CNN
- **Data Visualization & Analytics:** Power BI, Tableau, Excel
- **Cloud & Devops:** AWS (EC2, S3, IAM, Sagemaker), Docker, CI/CD, Jenkins, Git
- **Databases & Big Data:** MySQL, PostgreSQL, Hadoop, GitHub
- **CourseWork:** Probability & Statistics, Discrete mathematics, Data Structure and Algorithms, Deep Learning, Machine Learning, Cloud Computing, Data Science, Object-Oriented programming, Operating Systems

EXPERIENCE

Menzies Mission | Software Engineer, Machine Learning

Feb 2025- Dec 2025

- Improved NGO program efficiency by 30% by developing donor segmentation models (K-Means, Random Forest) in Python, Scikit-Learn, and TensorFlow, helping optimize fundraising campaigns and resource allocation.
- Predicted donation patterns using time-series forecasting (ARIMA, LSTM) to support budget planning and community program funding decisions.
- Deployed NLP pipelines with Hugging Face Transformers and LangChain to summarize beneficiary feedback and field reports, reducing manual review time by 40%.

Power Master Industries | Data Analyst

Sep 2021-July2022

- Improved production and reporting efficiency by around 25% by analyzing manufacturing and sales data to identify process delays and resource gaps.
- Developed interactive Power BI dashboards to visualize daily output, sales, and inventory status, aiding management in decision-making.
- Cleaned and standardized large datasets using Python (Pandas) and SQL, significantly increasing data accuracy.
- Collaborated with operations, quality, and sales teams to resolve data inconsistencies and deliver actionable insights.

ShreeHari Technology | Machine Learning Intern

June 2021-Aug2021

- Processed and analyzed client business datasets (customer transactions, usage logs, and service requests) using Python, Pandas, and NumPy, delivering insights through visual dashboards and charts for decision-making.
- Built Deep Neural Networks (DNN) to classify user behavior and predict which users might stop using the service, improving prediction accuracy by 15%.
- Developed a sentiment analysis module (Naïve Bayes, SVM) for a client feedback system, automatically classifying user comments to improve engagement and response efficiency.

ACADEMIC PROJECTS

Honeypot Attack Classification & Forecasting [Python, Scikit-Learn, ARIMA, LIME] | [Link](#)

- Built Random Forest and XGBoost classification models on honeypot attack data and evaluated performance using Classification Reports, ROC-AUC Curve, and Confusion Matrix to identify the best model.
- Applied ARIMA forecasting to predict future attack timing and protocols, and enhanced interpretability by integrating LIME (Explainable AI) to identify key features driving predictions.

Speech-to-Text Summarization [Python, TensorFlow, AWS SageMaker] | [Link](#)

- Reduced manual notetaking by 75% by building a speech-to-text summarization system that converts lecture audio into concise notes with 90% accuracy.
- Developed an end-to-end NLP pipeline using Google Speech-to-Text API and transformer-based summarization models and deployed it on AWS SageMaker for real-time processing and easy access.

Movie Recommendation System [Python, Pandas, Scikit-Learn, HTML, CSS, JavaScript] | [Link](#)

- Developed a content-based recommendation engine achieving 96% accuracy in suggesting relevant movies.
- Built a web-based interface with HTML, CSS, and JavaScript for user interaction and real-time movie recommendations.
- Integrated a SQL database to store movie metadata and user preferences efficiently.

CERTIFICATIONS

- AWS Certified Cloud Practitioner | [Link](#)
- Machine Learning Specialization | [Link](#)